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ST4005CEM Database System

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1. Create table customer with column name customer_id, customer_name, contact_number and country.

```
• create table cutomers(  
  customer_id int primary key ,  
  customer_name varchar(20),  
  contact_number char(10),  
  country varchar(20)  
);
```

2. Create table order with column name order_id, order_date, amount, customer_id (foreign key referencing customer)

```
• create table orders(  
  order_id int primary key ,  
  order_date date default(curdate()),  
  amount decimal(10,2),  
  customer_id int,  
  foreign key(customer_id) references cutomers(customer_id)  
);
```

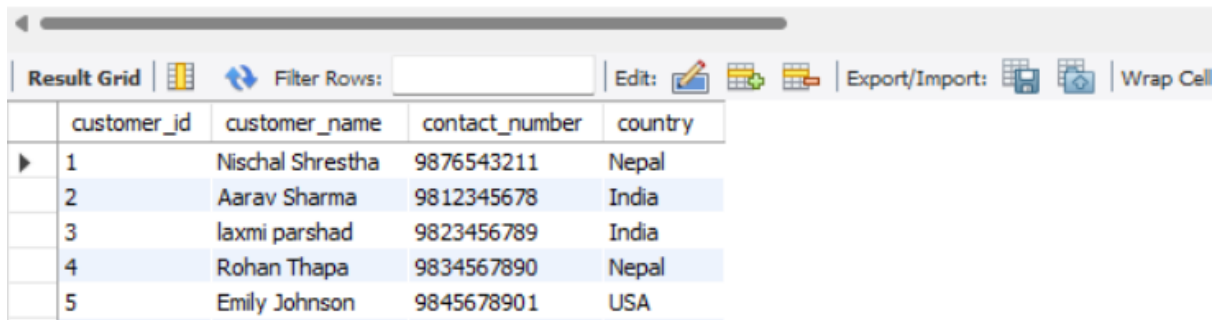
3. Insert a sample data minimum 5 records in both table.

```
INSERT INTO cutomers VALUES (1, "Nischal Shrestha", "9876543211", "Nepal");
INSERT INTO cutomers VALUES (2, "Aarav Sharma", "9812345678", "India");
INSERT INTO cutomers VALUES (3, "Priya Patel", "9823456789", "India");
INSERT INTO cutomers VALUES (4, "Rohan Thapa", "9834567890", "Nepal");
INSERT INTO cutomers VALUES (5, "Emily Johnson", "9845678901", "USA");
INSERT INTO cutomers VALUES (6, "Sita Gurung", "9856789012", "Nepal");
INSERT INTO cutomers VALUES (7, "James Carter", "9867890123", "UK");
INSERT INTO cutomers VALUES (8, "Anita Rai", "9878901234", "Nepal");
INSERT INTO cutomers VALUES (9, "Rahul Verma", "9889012345", "India");
INSERT INTO cutomers VALUES (10, "Sophia Williams", "9890123456", "Australia");
```

```
28 • INSERT INTO orders (order_id,amount,customer_id) VALUES
29 (1, 1500.00, 1),
30 (2, 2300.50, 3),
31 (3, 750.75, 5),
32 (4, 4200.00, 2),
33 (5, 980.25, 7),
34 (6, 3150.99, 4);
```

4. Update the contact name of customer with ID 3 to 'laxmi prasad'.

```
39 • update cutomers set customer_name ="laxmi parshad" where customer_id=3;
```



	customer_id	customer_name	contact_number	country
▶	1	Nischal Shrestha	9876543211	Nepal
	2	Aarav Sharma	9812345678	India
	3	laxmi parshad	9823456789	India
	4	Rohan Thapa	9834567890	Nepal
	5	Emily Johnson	9845678901	USA

5. Update the amount of order with ID 2 to 999.99.

```
40 • update orders set amount=999.99 where order_id=2;
```

Result Grid

	order_id	order_date	amount	customer_id
▶	1	2026-05-04	1500.00	1
	2	2026-05-04	999.99	3
	3	2026-05-04	750.75	5
	4	2026-05-04	4200.00	2
	5	2026-05-04	980.25	7

6. Delete the customer with customer_id = 5.

```
42 • delete from cutomers where customer_id=5;  
43 • select * from cutomers;
```

Result Grid

	customer_id	customer_name	contact_number	country
▶	1	Nischal Shrestha	9876543211	Nepal
	2	Aarav Sharma	9812345678	India
	3	laxmi parshad	9823456789	India
	4	Rohan Thapa	9834567890	Nepal
	6	Sita Gurung	9856789012	Nepal

7. Delete all orders with an amount less than 1000.

```
50 • delete from orders where amount<1000;
51 • select * from orders;
```

Result Grid				
Filter Rows:				
Edit:				
	order_id	order_date	amount	customer_id
▶	1	2026-05-04	1500.00	1
	4	2026-05-04	4200.00	2
	6	2026-05-04	3150.99	4
•	NULL	NULL	NULL	NULL

8. list all orders sorted by amount in descending order.

```
51 • select * from orders order by amount desc;
```

Result Grid				
Filter Rows:				
Edit:				
	order_id	order_date	amount	customer_id
▶	4	2026-05-04	4200.00	2
	6	2026-05-04	3150.99	4
	1	2026-05-04	1500.00	1
•	NULL	NULL	NULL	NULL

9. Display the top 5 most expensive orders.

```
49
50 • select * from orders order by amount desc limit 5;
```

Result Grid				
Filter Rows:				
Edit: Export				
	order_id	order_date	amount	customer_id
▶	4	2026-05-04	4200.00	2
	6	2026-05-04	3150.99	4
	2	2026-05-04	2300.50	1
	1	2026-05-04	1500.00	1
	5	2026-05-04	980.25	6

orders 47 ×

10. Show customer names sorted alphabetically.

```
51
52 • select * from cutomers order by customer_name;
```

Result Grid | Filter Rows: Edit: E

	customer_id	customer_name	contact_number	country
▶	2	Aarav Sharma	9812345678	India
	8	Anita Rai	9878901234	Nepal
	7	James Carter	9867890123	UK
	3	laxmi parshad	9823456789	India
	1	Nischal Shrestha	9876543211	Nepal
	9	Rahul Verma	9889012345	India
	4	Rohan Thapa	9834567890	Nepal
	6	Sita Gurung	9856789012	Nepal
	10	Sophia Williams	9890123456	Australia
★	NULL	NULL	NULL	NULL

11. Get the 3 most recent orders by order_date.

```
56 • select * from orders order by order_date desc limit 3;
57
```

Result Grid | Filter Rows: Edit: Export/Imp

	order_id	order_date	amount	customer_id
▶	8	2026-05-06	2300.30	1
	1	2026-05-04	1500.00	1
	2	2026-05-04	2300.50	1
★	NULL	NULL	NULL	NULL

12. Retrieve orders, skipping the first 2 records and showing the next 5.

```
58 • select * from orders limit 2,5;
```

59

order_id	order_date	amount	customer_id
3	2026-05-04	750.75	3
4	2026-05-04	4200.00	2
5	2026-05-04	980.25	6
6	2026-05-04	3150.99	4
7	2026-04-06	2300.30	1
NULL	NULL	NULL	NULL

13. Find all customers whose name starts with 'A'.

```
59 • select * from cutomers where customer_name like 'a%';
```

60

customer_id	customer_name	contact_number	country
2	Aarav Sharma	9812345678	India
8	Anita Rai	9878901234	Nepal
NULL	NULL	NULL	NULL

14. Find customers whose contact name ends with 'son'.

```

59 • select * from cutomers where customer_name like 'a%';
60 • select * from cutomers where customer_name like '%son';
61

```

Result Grid	Filter Rows:	Edit:	Export/Import:
customer_id	customer_name	contact_number	country
11	Ramson	9876543456	China
NULL	NULL	NULL	NULL

15. List orders where amount is between 500 and 1500.

```

63 • select * from orders where amount between 500 and 1500;
64

```

Result Grid	Filter Rows:	Edit:	Export/Import:
order_id	order_date	amount	customer_id
1	2026-05-04	1500.00	1
3	2026-05-04	750.75	3
5	2026-05-04	980.25	6
NULL	NULL	NULL	NULL

16. List customers from countries 'USA', 'Canada', and 'UK' .

```

65 • select * from cutomers where country in ("USA","Canada","uk");
66

```

Result Grid	Filter Rows:	Edit:	Export/Import:
customer_id	customer_name	contact_number	country
7	James Carter	9867890123	UK
NULL	NULL	NULL	NULL

17. List all orders with their amount and a status column showing 'High' if amount > 1000, otherwise 'Low'.

```
66 • select * , case
67   when amount > 1000 then 'high'
68   else 'low'
69   end as status from orders;
```

Result Grid   Filter Rows: Export: 					
	order_id	order_date	amount	customer_id	status
▶	1	2026-05-04	1500.00	1	high
	2	2026-05-04	2300.50	1	high
	3	2026-05-04	750.75	3	low
	4	2026-05-04	4200.00	2	high
	5	2026-05-04	980.25	6	low
	6	2026-05-04	3150.99	4	high
	7	2026-04-06	2300.30	1	high
	8	2026-05-06	2300.30	1	high

18. Demonstrate what happens to orders when a customer is deleted using ON DELETE CASCADE.

Output				
Action Output				
#	Time	Action	Message	
39	18:59:39	delete from orders where amount<1000	Error Code: 1175. Y	
40	18:59:44	set sql_safe_updates=0	0 row(s) affected	
41	19:00:00	delete from orders where amount<1000	2 row(s) affected	

```
47
48 • alter table orders drop constraint new_foreign_key;
49 • alter table orders add constraint new_foreign_key
50   foreign key(customer_id) references customers(customer_id) on delete set null;
51 • select * from orders;
52 • delete from orders where amount<1000;
53 • set sql safe updates=0;
```

Output			
Action Output			
#	Time	Action	Message
✓ 44	19:01:39	alter table orders add constraint new_foreign_key foreign key(customer_id) references customers(customer_id) o...	6 row(s) affected Records
✓ 45	19:01:52	select * from orders LIMIT 0, 1000	6 row(s) returned
✓ 46	19:02:05	delete from orders where amount<1000	0 row(s) affected

20. Write an SQL query to display all customers and their corresponding orders.

```

75     -- join as status from orders;
76 • select * from cutomers inner join orders on cutomers.customer_id=orders.customer_id;
77

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

IA

	customer_id	customer_name	contact_number	country	order_id	order_date	amount	customer_id
▶	1	Nischal Shrestha	9876543211	Nepal	1	2026-05-04	1500.00	1
	1	Nischal Shrestha	9876543211	Nepal	2	2026-05-04	2300.50	1
	2	Aarav Sharma	9812345678	India	4	2026-05-04	4200.00	2
	4	Rohan Thapa	9834567890	Nepal	6	2026-05-04	3150.99	4
	1	Nischal Shrestha	9876543211	Nepal	7	2026-04-06	2300.30	1
	1	Nischal Shrestha	9876543211	Nepal	8	2026-05-06	2300.30	1

21. Write an SQL query to list all customers and their orders, including customers who haven't placed any orders.

```

76
77 • select * from cutomers left join orders on cutomers.customer_id=orders.customer_id;
78

```

Result Grid		Filter Rows:		Export:	Wrap Cell Content:			
	customer_id	customer_name	contact_number	country	order_id	order_date	amount	customer_id
▶	1	Nischal Shrestha	9876543211	Nepal	8	2026-05-06	2300.30	1
	1	Nischal Shrestha	9876543211	Nepal	7	2026-04-06	2300.30	1
	1	Nischal Shrestha	9876543211	Nepal	2	2026-05-04	2300.50	1
	1	Nischal Shrestha	9876543211	Nepal	1	2026-05-04	1500.00	1
	2	Aarav Sharma	9812345678	India	4	2026-05-04	4200.00	2
	3	Iaxmi parshad	9823456789	India	NULL	NULL	NULL	NULL
	4	Rohan Thapa	9834567890	Nepal	6	2026-05-04	3150.99	4
	6	Sita Gurung	9856789012	Nepal	NULL	NULL	NULL	NULL
	7	James Carter	9867890123	UK	NULL	NULL	NULL	NULL
	8	Anita Rai	9878901234	Nepal	NULL	NULL	NULL	NULL
	9	Rahul Verma	9889012345	India	NULL	NULL	NULL	NULL
	10	Sophia Williams	9890123456	Australia	NULL	NULL	NULL	NULL
	11	Ramson	9876543456	China	NULL	NULL	NULL	NULL

22. Write an SQL query to display Order ID, Order Date, and the Customer Name for each order.

```

79
80 • select order_id,order_date,customer_name from orders join cutomers on orders.customer_id=cutomers.customer_id;
81

```

order_id	order_date	customer_name
1	2026-05-04	Nischal Shrestha
2	2026-05-04	Nischal Shrestha
4	2026-05-04	Aarav Sharma
6	2026-05-04	Rohan Thapa
7	2026-04-06	Nischal Shrestha
8	2026-05-06	Nischal Shrestha

23. Display customer names and order amounts sorted by customer name.

```

81 • select customer_name,amount from cutomers
82 join orders on cutomers.customer_id=orders.customer_id
83 order by customer_name;
84

```

customer_name	amount
Aarav Sharma	4200.00
Nischal Shrestha	1500.00
Nischal Shrestha	2300.50
Nischal Shrestha	2300.30
Nischal Shrestha	2300.30
Rohan Thapa	3150.99

24. Use INNER JOIN to list customers who have placed orders with an amount greater than 1000.

84

85 • `select * from cutomers inner join`

86 `orders on cutomers.customer_id=orders.customer_id`

87 `where amount>1000;`

88

Result Grid		Filter Rows:		Export:	Wrap Cell Content: IA			
	customer_id	customer_name	contact_number	country	order_id	order_date	amount	customer_id
▶	1	Nischal Shrestha	9876543211	Nepal	1	2026-05-04	1500.00	1
	1	Nischal Shrestha	9876543211	Nepal	2	2026-05-04	2300.50	1
	2	Aarav Sharma	9812345678	India	4	2026-05-04	4200.00	2
	4	Rohan Thapa	9834567890	Nepal	6	2026-05-04	3150.99	4
	1	Nischal Shrestha	9876543211	Nepal	7	2026-04-06	2300.30	1
	1	Nischal Shrestha	9876543211	Nepal	8	2026-05-06	2300.30	1